

Alemu Sisay Nigru

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AI Research Scientist | PhD, AI in Medicine

PhD Researcher at the University of Brescia with specialized expertise in Deep Learning for Medical Imaging and Clinical Decision Support. Proven track record in developing MRI severity grading algorithms and fine-tuning Large Language Models (LLMs) for radiological report generation. Experienced in cross-continental clinical collaboration at **NYU** and leading large academic teams.

Experience

Lead AI Engineer | Xray Service, Brescia, Italy

December 2025 – Now

- **Achieved a 75–85% reduction in active reporting time** for senior radiologists by developing and deploying **LumbarAI**, a Swin3D Transformer-based framework for automated MRI analysis.
- **Reduced physician data-entry latency** by automating the grading of 11 radiological features and integrating **PACS/DICOM workflows** for seamless image retrieval.
- **Ensured 100% clinician oversight** by engineering a **Human-in-the-Loop** framework with real-time explainability heatmaps (XAI).
- **Facilitated immediate clinical documentation** across regions by building a bilingual (EN/IT) report generation engine.

Visiting PhD researcher 2 | New York University (NYU)

April 2025 – October 2025

- **Optimized diagnostic accuracy** for Low Back Pain (LBP) severity grading by fine-tuning LLMs for automated, human-like radiological report generation.
- **Developed & Optimized** machine learning algorithms for LBP (Low Back Pain) severity grading, achieving robust performance across diverse MRI datasets.
- **Colaborated** with multidisciplinary teams on imaging pipelines and model interpretability.

Visiting PhD researcher 1 | Center for Digital Health & Implementation Science (CDHI), Ethiopia

September 2024 – December 2024

- **Integrated AI into EMR:** Contributed to the design and deployment of AI diagnostic tools within Electronic Medical Record (EMR) systems.
- **UI/UX Development:** Created graphical interfaces to facilitate the seamless adoption of Deep Learning models by physicians in routine medical activities.

Technical Skills

- **AI, Machine Learning, & Deep Learning:** PyTorch, TensorFlow, Swin3D Transformers, LLMs, Transfer Learning, Domain Adaptation, Explainable AI (XAI).
- **Medical Computing:** DICOM, PACS, NIfTI, Nibabel, Nipype, MRI/Neuroimage Processing.
- **Software & Deployment:** Python (Advanced), C++, SQL, ETL Pipelines, Human-in-the-Loop Frameworks, UI/UX (Streamlit/Gradio).
- **Clinical Research:** Statistical Methods, MRI Severity Grading, Patient-Reported Outcomes, External Validation.

Education

Ph.D. in Artificial Intelligence in Medicine

University of Brescia, Italy | 2022 – 2025 (Expected)

Dissertation: [AI-driven clinical decision support for personalized management of low back pain.](#)

M.Sc. in Communication Technologies and Multimedia

University of Brescia, Italy | 2019 – 2022

Final Grade: 110 Cum Laude/110

Thesis: [Deep learning-based neuroimage retrieval](#)

B.Sc. in Electrical and Computer Engineering

University of Gondar, Ethiopia | 2012 – 2016

Final Grade: 3.89/4.00 — **Gold Medalist**

Thesis: [Fingerprint-based Attendance System \(FAS\) using ASP.NET and C#](#)

Publications:

Peer-Reviewed:

Alemu Sisay Nigru, Sergio Benini, Matteo Bonetti, Graziella Bragaglio, Michele Frigerio, Federico Maffezzoni, Riccardo Leonardi “External validation of SpineNetV2 on a comprehensive set of radiological features for grading lumbosacral disc pathologies.” **North American Spine Society Journal (NASSJ)**, 2024.

Under Review / Preprints

- *Alemu Sisay Nigru, Sergio Benini, Yao Wang, Federico Maffezzoni, Matteo Bonetti, Michele Frigerio, Graziella Bragaglio, Serena Miglio, Riccardo Leonardi* “Toward a Clinically Integrated AI Framework for Personalized Spine Care.” SSRN preprint, 2026.
- *Alemu Sisay Nigru, Michele Svanera, Austin Dibble, Connor Dalby, Mattia Savardi, Sergio Benini* Rewiring Development: Leveraging Adult Brain Priors for Enhancing Infant Brain MRI Segmentation. arXiv preprint, 2025.
- *Alemu Sisay Nigru, , Riccardo Leonardi, Yao Wang, Federico Maffezzoni, Serena Miglio, Matteo Bonetti, Michele Frigerio, Graziella Bragaglio, Sergio Benini* “LumbarAI: A Swin3D Transformer-Based Framework for Computer-Aided Diagnosis of Lumbar Spine Disorders from MRI” (*under review*).
- *Alemu Sisay Nigru, Sergio Benini, Yao Wang, Federico Maffezzoni, Matteo Bonetti, Michele Frigerio, Graziella Bragaglio, Serena Miglio, Riccardo Leonardi* “Unsupervised Clustering of Psychosocial Profiles for Patient Stratification in Spinal Pain Management” (*under review*).

Certificates & Trainings:

- Oxford Machine Learning Summer School — AI for Global Goals, University of Oxford
- Applied Data Science Lab — WorldQuant University
- Higher Diploma Program Certificate in Teaching Methods — University of Gondar
- Several additional short course certificates in ML, Deep Learning, and AI.